A Secure Role-Based Access Control Framework for Employee Plantilla Management System in the Local Government Unit of Magallanes, Agusan del Norte

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Abstract— **This study addresses the security weaknesses in the Excel-based employee plantilla management system of the Local Government Unit of Magallanes, Agusan del Norte. The existing system lacked proper access control, exposing sensitive employee data to unauthorized access and non-compliance with the article III, section 11 and section 12 of the Data Privacy Act of 2012. The study aims to developed a secure role-based access control framework to regulate user privileges, enhance data security, and ensure compliance with the Data Privacy Act. Additionally, it includes user activity logging and a lockout mechanism after multiple failed logins attempts to enhance accountability, traceability, and monitoring of actions performed by the users. A waterfall model covers planning, analysis, design, development, implementation, and testing phases. The system was developed using PHP, the Laravel framework, and MySQL. The ISO/IEC 25010 standard evaluated software quality attributes such as performance, security, usability, confidentiality, reliability, and user satisfaction. The system achieved high acceptance, with overall user satisfaction score of 4.65. It significantly improved the security, operational efficiency, and compliance. Feature like role-based access control, activity logging, and the lockout mechanism were well-received. The system enhances the LGU's ability to manage employee plantilla records securely and serves as a model for future digital transformation and compliance in the local government human resource management.**

Keywords— Role-Based Access Control, Plantilla, Information Security, Data Privacy Act of 2012

I. Introduction

Within the context of government and public administration in the Philippines, the term "plantilla" relates to the authorized master list of permanent positions in government offices and Local Government Units. This list assists the government human resource management office in the systematic allocation of budgets, planning salaries, and managing benefits, promotions, retirements, and other movements of personnel [1]. This list can assist the human resource office in geospatial planning and the systematic breakdown of the office's human capital for work expansion and the government of public employees. This serves the public and the government agencies in planning and streamlining control. This justifies the economic allocation of seats in the government for public accountability.

In today's digital world, Local Government Units (LGUs) increasingly turn to computerized systems to handle administrative task, including keeping track of employee plantilla records [2]. These records contain essential information such as employee status, job classification, salary grades, hire dates, and plantilla items numbers, all vital for making informed decisions and efficiently managing resources. However, with this growing dependence on digital systems comes a rising concerns about data privacy, integrity, and risk of unauthorized access, particularly in human resource offices that handle sensitive information [3].

Despite advancements, the local government unit of Magallanes still has not embraced in digital transformation for its human resource services offices and still uses Microsoft Excel-based record-keeping. These record-keeping methods do not have adequate privacy protection. The lack of finely-tuned access control and in the current system puts critical information at risk from both internal and external sources. Traditional access control systems provide vastly insufficient systems to prevent the unauthorized changes and lack of accountability which greatly increases the risk of exposure and breaches. This increases the risks of non-compliance with internal information security control policies and external government data protection laws like Republic Act No. 10173, otherwise known as the Data Privacy Act of 2012 [4].

Although several digital solutions have been proposed for data management in government offices, the existing systems, such as the one used by the LGU of Magallanes, fail to implement sufficient access control and data protection

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features. Traditional systems do not adequately restrict user access based on their roles and responsibility and prevent unauthorized access. This gap in security is a critical concern for LGU's which protect sensitive employee information while also ensuring operational efficiency.

This study addresses the identified challenges by developing a "Secure Role-based Access Control Framework for Employee Plantilla Management Systems" specially designed for the Local Government Unit of Magallanes, Agusan del Norte. This system is capable of constraining access of the user based on defined organizational roles and responsibilities, lessening the danger of unauthorized data access, and maintaining operational integrity. Furthermore, the system implements user activity logging to enhance accountability, traceability, and monitoring of actions performed on sensitive personnel records. It also includes a security feature that disables the login interface after three consecutive failed login attempts and enforce a 25-second lockout period before allowing re-access helping deter unauthorized hacking attempts [5].

This paper is organized as follows: Section II presents the methodology employed in the system's development, including the design, development, and implementation phases. Section III presents the evaluation results, including system quality features such as performance efficiency, security, usability, reliability, and user satisfaction. Finally, Section IV concludes the paper, summarizing the findings and proposing the future directions for improvement in the Human Resources Management Office of the Local Government Unit of Magallanes, Agusan del Norte.

II. METHODOLOGY

This chapter dives into the project design, highlighting how a developmental research approach proves effective for this study. It serves as a guide for the system designer to gather and tackle development requirements. The assessment of the system follows the ISO/IEC 25010 framework, with emphasis on software quality characteristics including efficiency, security, usability, confidentiality, integrity, accountability and user satisfaction. This provides a clear framework for assessing and improving performance.[6].

A. System Design and Development

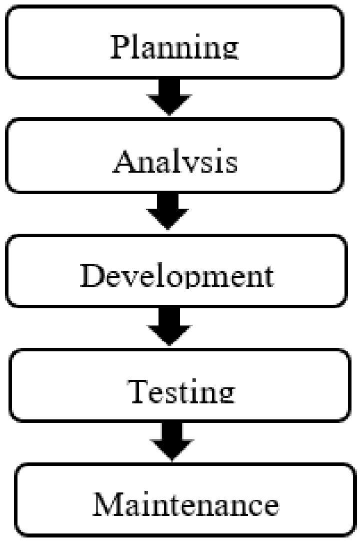

Fig. 1. Process Model of the Secure Role-based Access Control Framework for Employee Plantilla Management System

Fig. 1 shows the process model of the secure role-based access control framework for employment plantilla management system. First, the planning stage began with an in-depth collection of data through a series of interviews with the staff from the Human Resource Management Office at the Local Government Unit of Magallanes, Agusan del Norte. The information gathered is aimed at truly understanding the needs of the users, pinpointing key requirements, and identifying specific issues. It also sets clear goals, figures out the essential resources needed for the project's success, and incorporates the necessary security measures. This phase laid the groundwork for the design and development of the system, making sure that the project's scope aligns with user expectations.

Second, is the analysis phase, where we pinpoint the processes of the current system as well as its constraints and limitation. These elements become system models, and includes a data flow, a use-case and an architecture of the system. A logical model in the form of a Data Flow Diagram (DFD) depicts the flows of the system information. To represent user interaction more vividly, the developers made a use-case diagram that describes the user's system engagements and the features the system offers. The system architecture captures all the essentials and high-level constituents of the system, while outlining its structure, operation, and the primary mechanisms. This model significantly helps in identifying the processes and interchanged data and the fundamental components of the system.

Third is the development phase, in which the front-end was built using the Laravel web application framework and Hyper Preprocessor (PHP) programming language, while MySQL was used as the data storage and management system for the back-end. System reviews were conducted, and in preparation for implementation at the Local Government Unit of Magallanes, testing was done along with the feedback gathered from the Human resource Management Office

Fourth is the system phase. Before rolling out the system, a system test was carried out under the ISO/IEC standard, which evaluates system quality attributes: performance, efficiency, security, usability, confidentiality, integrity, accountability, and satisfaction. The system performed admirably, scoring 4.65 overall satisfaction. The findings underscore the effectiveness of the approach: the role-based access control not only improved the safeguards placed on the system, but also improved the efficiency of operational tasks. With the role-base access control system, the LGU can now securely and efficiently handle employee plantilla records. The system was successfully deployed at the Local Government Unit of Magallanes, which include installing it on the server and configuring it for live operation. This deployment made the system accessible to users, allowing them to effectively monitor and manage its functions.

Lastly, the maintenance phase kicked off right after deployment, with a keen eye on how the system performed in real-world scenarios. Regular check-ups were carried out to make sure everything was running smoothly, just as it was meant to. This phase also included making updates and improvements based on user feedback.

B. Framework and Models

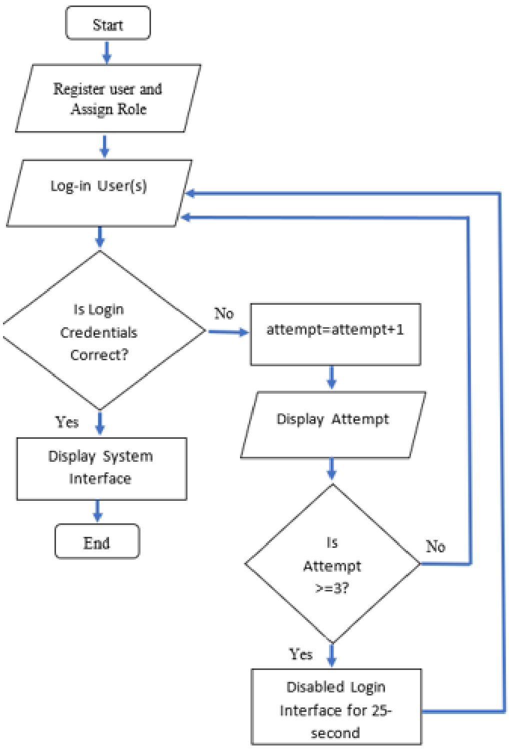

Fig.2. Flowchart of a Role-Based Access Framework for Plantilla Management System of the LGU of Magallanes, Agusan del Norte

Fig. 2 shows the conceptual model highlights the login authentications process, lockout mechanism, and role assignment as part of the study's contributions to information security design. It starts with the Human Resource Management Head registering users and assigning them a specific role. After registration, users proceed to the login interface to input their identifications. The system then validates whether those credentials are correct. If it is correct, the user gains access and the system interface appear based on the assigned role, wrapping up the process. On the other hand, if the credentials are incorrect, the system counts the failed login attempts and shows how many tries have been made. It then checks if the number of failed attempts has hit three or more. If not, the user is sent back to try logging in again. However, if the failed attempts reach three or higher, the system temporarily locks the login interface for 25 seconds to thwart any brute-force attacks. Once that brief lockout is over, the user can give logging in another shot. This approach boosts system security by limiting unauthorized access through repeated login failures.

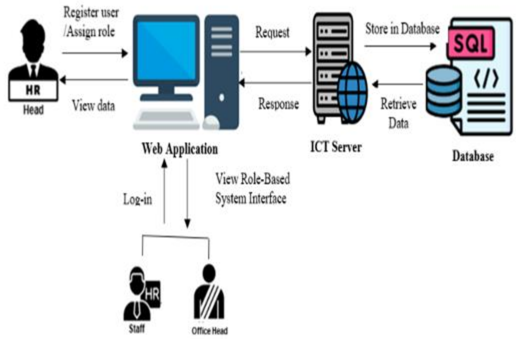

Fig.3. System Architecture of the Role-Based Access Control Framework for Plantilla Management System

Fig. 3 Illustrates the overall framework contribution, showing how different user roles interact with the ICT server and database to ensure secure and structured access control. At the top level, the Human Resource Head is responsible for registering users and assigning roles through the web application interface. Staff and Office Heads log into the system, where they encounter a role-based interface designed to match their specific access levels. User requests are sent to the ICT Server, which serves as middleware, processing these requests and communicating with the Database. The database, managed via SQL, efficiently stores and retrieves user data based on system transactions. Responses from the database are sent back to the web application through the ICT server, allowing for real-time interaction. Additionally, the Human Resource Head has the authority to view data records, which aids in oversight and management. This architecture highlights the importance of secure, role-based access control and effective data management through centralized processing.

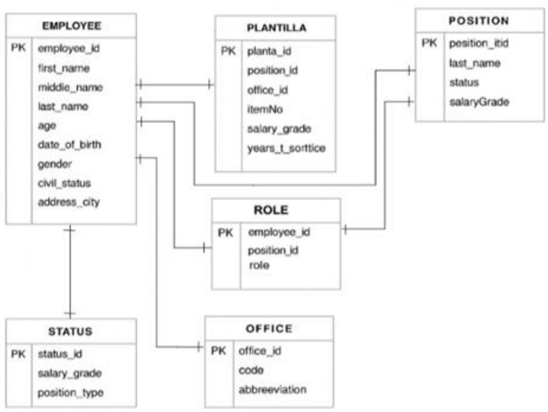

Fig.4. Entity Relationship Diagram of the Role-Based Access Control Framework for Plantilla Management System

Fig. 4 represents the conceptual data model contribution, defining entities and relationships that enforce secure role-based interactions for employee plantilla records. It highlights seven key entities: EMPLOYEE, PLANTILLA, POSITION, ROLE, STATUS, and OFFICE. The EMPLOYEE entity holds personal information, using employee_id as the primary key, and connects to STATUS, ROLE, and PLANTILLA. PLANTILLA records job assignments, with planta_id serving as the primary key, and establishes a many-to-one relationship with POSITION and OFFICE. POSITION outlines organizational roles and links to both PLANTILLA and ROLE. ROLE creates a many-to-many relationship between EMPLOYEE and POSITION through a composite key, which also includes a role attribute. STATUS categorizes employment types and connects back to EMPLOYEE. Finally, OFFICE represents different departments, identified by office_id, and links to both PLANTILLA and STATUS to illustrate the organizational structure.

III. Results and Discussions

This section highlights the results of the system, showcasing application screenshots of the user interface along with the assessment and validation of users for the Secure Role-Based Access Framework for Employee Plantilla Management System in Municipality of Magallanes, Agusan del Norte.

A. Implementation (System Interface)

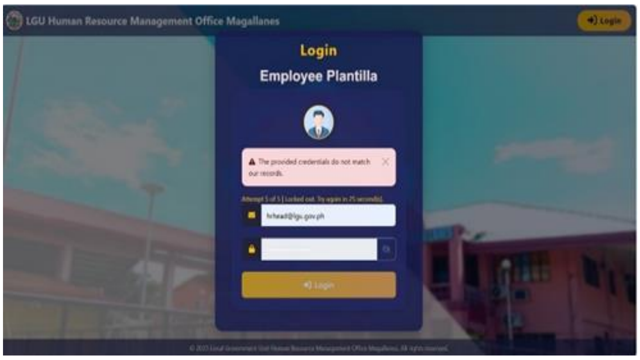

Fig. 5. Log-in Page of the implemented system

Fig. 5 demonstrate the system implementation of secure login with a lockout feature after multiple failed attempts. This interface will display when the users will open the system. In this interface, the users will enter their users name and password to get inside the dashboard of the system. If they are new on this platform, the system will suggest to register first to be able to use the system. This interface is designed to automatically disable after three consecutive failed login attempts. It will enforce a 25-second lockout period before re-enabling access.

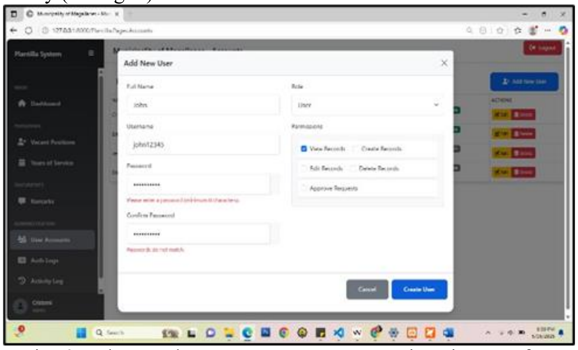

Fig. 6. Role-Based User Account System Registration Interface

Fig. 6 illustrates the practical implementation of the Role-based Access Control, where HR Head assigns specific roles and permissions to users.

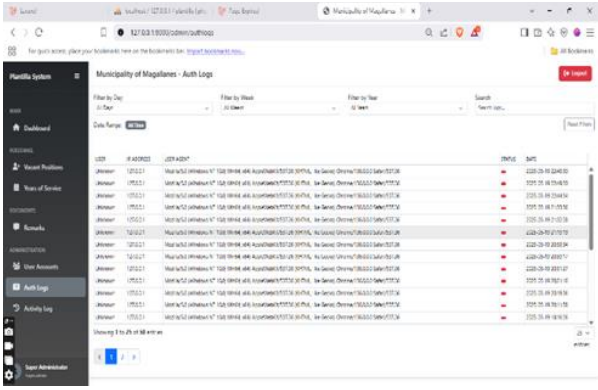

Fig. 7. Authentication Logs System Interface

Fig. 7 demonstrates the system's implementation feature for tracking login attempts, ensuring accountability and monitoring of user activities. In this interface, all login and authentication related activities within the system are recorded and monitor.

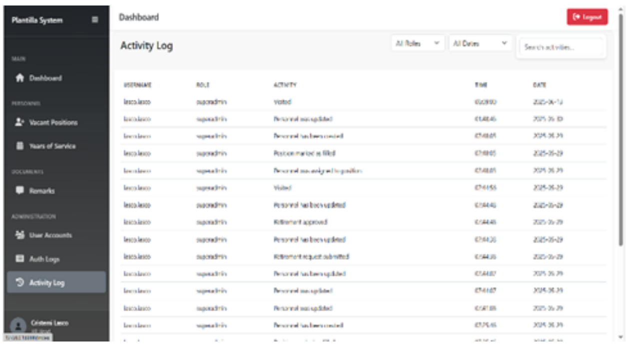

Fig. 8. Activity Logs System Interface

Fig. 8 shows how the system records and tracks user actions by logging who did what and when to strengthen auditability and operational transparency.

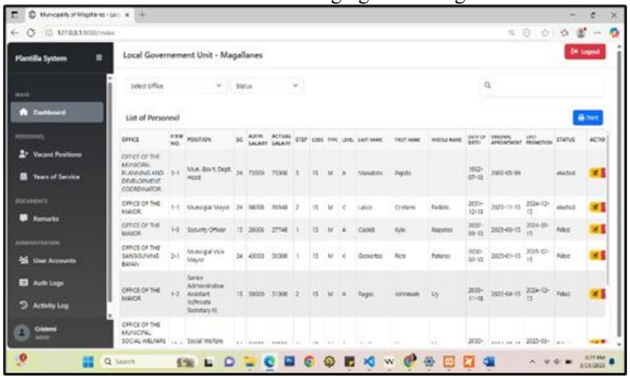

Fig. 9. System Dashboard Interface

Fig. 9 shows the system dashboard interface for users. It contains details list summary of personnel and its corresponding details per office. It has also a drop-down menu to filter the desired data to be display on the interface.

B. *Evaluation and Results*

This phase involved the evaluation of the system by key stakeholders, including the HR head, HR Staff, and other office heads and personnel. The developers provided evaluators with access to the system and a standardized questionnaire designed to assess key quality attributes such as performance efficiency, security, usability, confidentiality, integrity, accountability, and user satisfaction. The system underwent a one-week testing period to validate its operational functionality under real-world conditions. The questionnaire used a scale of one (1) to five (5), where 1 indicated the lowest and five the highest level of performance. Each rating was associated with a specific performance level and interval, as the accompanying table details. The assessment criteria were aligned with the ISO/IEC 25010 standard for product quality evaluation.

a) Rating Scale: Table I shows the rating scale of the system's performance evaluated on a weighted scale with five levels. A score between 4.21 and 5.0 indicated a High Acceptable performance, while score from 3.41 to 4. 20 are considered Acceptable. A performance score between 2.61 and 3.40 falls into the Moderately Acceptable range. Scores between 1.81 and 2.60 are deemed Fairly Acceptable, and scores from 1.00 to 1.80 are classified as Not Acceptable.

TABLE I. RATING SCALE

Weight	Scale of Mean	Performance Level
5	4.21-5.00	Highly Acceptable
4	3.41-4.20	Acceptable
3	2.61-3.40	Moderately Acceptable
2	1.81-2.60	Fairly Acceptable
1	1.00-1.80	Not Acceptable

b) Performance Efficiency: Table II evaluates the performance efficiency of a secure role-based access framework. The overall average score in this area is 4.62, with a standard deviation of 0.48, which falls under the category of Highly Acceptable. The standout aspect of performance efficiency is item 2, "Resource utilization," which received a mean score of 4.72 and a standard deviation of 0.45, also rated as Highly Acceptable. On the other hand, the item that received the lowest marks is item 1, "Time behavior," with a mean score of 4.45 and a standard deviation of 0.50, yet still rated as Highly Acceptable. These findings indicate that the role-based access control approach for enhancing information security in the employee plantilla management system is performing excellently, showcasing a system that is reliable, fast, secure, and consistently optimized.

TABLE II. PERFORMANCE EFFICIENCY

Performance Efficiency	Mean ($\bar{x}$)	Std. Deviation	Qualitative Description
1.Time behavior	4.54	0.50	Highly Acceptable
2.Resource utilization	4.72	0.45	Highly Acceptable
3.Capacity	4.60	0.49	Highly Acceptable
Overall mean	**4.62**	**0.48**	**Highly Acceptable**

c) System Security: Table III evaluates the system security level of a role-based access control framework aimed at enhancing information security within the Employee Plantilla Management System. The overall average score for this area is 4.56, with a standard deviation of 0.49, which falls under the category of Highly Acceptable. The standout item for security is item 4, "Accountability," boasting a mean score of 4.66 and a standard deviation of 0.48, also rated as Highly Acceptable. On the other hand, the item that received the lowest marks is item 3, "non-repudiation," which still holds a Highly Acceptable rating with a mean of 4.42 and a standard deviation of 0.50. These results indicate that the role-based access control framework is effectively strengthening information security in the system.

TABLE III. SYSTEM SECURITY

Security	Mean ($\bar{x}$)	Std. Deviation	Qualitative Description
1.Confidentiality	4.60	0.49	Highly Acceptable
2.Integrity	4.58	0.50	Highly Acceptable
3.non-repudiation	4.42	0.50	Highly Acceptable
4.Accountability	4.66	0.48	Highly Acceptable
5.Authenticity	4.52	0.50	Highly Acceptable
Overall mean	**4.56**	**0.49**	**Highly Acceptable**

d) System Usability: Table IV showcases the usability of the six criteria the appropriateness, recognizability, learnability, operability, user error protection, interface aesthetics and accessibility were all of them rated "Acceptable". Operability scored the highest (4.03), followed by learnability (4.00) and accessibility (3.97), while appropriateness recognizability had the lowest score (3.77). With an overall mean of 3.97, the results indicate that the system is user-friendly, easy to learn, and meets key usability standards.

TABLE IV. SYSTEM USABILITY

Usability	Mean (x̄)	Std. Deviation	Qualitative Description
1.Appropriateness recognizability	3.77	0.68	Acceptable
2.Learnability	4.00	0.69	Acceptable
3.Operability	4.03	0.56	Acceptable
4.User error protection	3.83	0.65	Acceptable
5.User interface aesthetics	3.87	0.51	Acceptable
6.Accessibility	3.97	0.56	Acceptable
Overall mean	**3.91**	**0.61**	**Acceptable**

e) System Reliability: Table V showcases the evaluation results for system reliability, focusing on four essential attributes: maturity, availability, fault tolerance, and recoverability. Each of these criteria was assessed using a five-point scale, and all components earned a glowing description of "Highly Acceptable." Availability stood out with the highest mean score of 4.68, suggesting that users found the system to be consistently accessible and operational. Recoverability wasn't far behind, with a mean score of 4.60, highlighting the system's impressive ability to bounce back after a failure. Maturity and fault tolerance also received commendable ratings, with mean scores of 4.58 and 4.52 respectively, indicating the system's stable performance over time and its knack for managing unexpected errors without causing major disruptions. Overall, the mean score for system reliability was 4.60, with a standard deviation of 0.49, reflecting a high and consistent level of user satisfaction. These findings suggest that the system delivers the high standards of reliability across all evaluated aspects.

TABLE V. SYSTEM RELIABILITY

Reliability	Mean (x̄)	Std. Deviation	Qualitative Description
1.Maturity	4.58	0.50	Highly Acceptable
2.Availability	4.68	0.47	Highly Acceptable
3.Fault tolerance	4.52	0.50	Highly Acceptable
4.Recoverability	4.60	0.49	Highly Acceptable
Overall mean	**4.60**	**0.49**	**Highly Acceptable**

f) Over-All User's Satisfaction: Table VI Overall satisfaction of the respondents with regards to the performance of the system. It obtained a mean rating of 4.65, it manifested that the majority of the respondents rated highly acceptable.

TABLE VI. OVER-ALL USER'S SATISFACTION

User's Satisfaction	Mean (x̄)	Std. Deviation	Qualitative Description
My overall satisfaction with the system.	4.64	0.48	Highly Acceptable
2.The overall quality of the system	4.66	0.48	Highly Acceptable
	4.65	**0.48**	**Highly Acceptable**

IV. CONCLUSION AND RECOMMENDATION

A. Conclusion

The system showed impressive performance during testing and met all the necessary specifications. The feedback from the HR head, staff, and other office leaders was overwhelmingly positive, with an overall user satisfaction rating of 4.65. The system excelled in several key areas, earning a performance efficiency score of 4.62, a usability score of 3.91, a security score of 4.56, and a reliability score of 4.60, all highlighting its high-level performance. Beyond these results, this study advances the application of role-based access control (RBAC) by demonstrating a practical, and regulatory-compliant framework tailored for plantilla management in a Local Government Unit. The system integrates unique enhancements including automatic login lockouts after failed attempts, dual logging mechanism for accountability, and ISO/IEC 25010-based evaluation ensuring both strong security and compliance with RA 1173.

Overall, the developed system not only improves user experience through its secure login functionality, role-based dashboards, and intuitive interface but also offers replicable model for secure digital transformation in local government unit human resource management.

B. Recommendation

The developers recommend that the future developers consider enhancing the system by incorporating the one-time password feature and the artificial intelligent anomaly detection to add extra layer of protection, making it significantly harder for hackers to exploit the system even if they manage to obtain a user's regular password.

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